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1. Introduction

1.1 Purpose

The purpose of the Information System for Mart is to develop a modern and efficient web-based retail management system that helps shop owners manage inventory, sales, product organization, and profit tracking digitally. The system is designed to reduce manual work, improve operational efficiency, and provide accurate business monitoring.

The platform combines inventory management, product searching, sales recording, and business analytics into a single centralized system. It also improves product accessibility through rack and shelf mapping and ensures secure user access through authentication and role-based permissions.

The project aims to provide a practical and scalable solution for small and medium retail businesses.

1.2 Scope

The scope of the project includes the design and development of a web-based system capable of handling the core operations of a retail store.

The system will provide:

- Product inventory management
- Product categorization and organization
- Rack and shelf mapping system
- Product searching using SKU and product name
- Sales transaction recording
- Real-time inventory updates
- Profit calculation and analytics
- User authentication and role management
- Dashboard and reporting features

The project is focused on improving retail management efficiency while maintaining accuracy, reliability, and ease of use.

2. Software Development Model

2.1 Prototyping Model

The Prototyping Model is a software development methodology in which an initial version of the software is developed quickly to understand system requirements and user expectations. This prototype is shown to users for evaluation and feedback. Based on the feedback, the prototype is refined repeatedly until the final system is developed.

Unlike traditional development approaches where requirements are fixed from the beginning, the prototyping model allows continuous interaction between developers and users during the development process.

This model is highly suitable for projects where user interface design, usability, and feature requirements may evolve during development.

2.2 Why Prototyping Model is Used

The Information System for Mart involves multiple user interactions such as inventory management, sales processing, product searching, and dashboard monitoring. Since user requirements may change after visualizing the system, the prototyping model is considered the most suitable development approach.

The main reasons for choosing the prototyping model are:

- It allows early visualization of the system
- Users can provide feedback during development
- Changes can be implemented easily
- It helps identify missing requirements early
- It reduces development risk and misunderstandings
- It improves overall usability and user satisfaction

Using this model helps ensure that the final system meets the practical needs of store owners and workers.

2.3 Phases of Prototyping Model

The development process using the prototyping model consists of the following phases:

1. Requirement Gathering

Initial system requirements are collected from users and stakeholders.

2. Quick Design

A basic design of the system interface and structure is prepared.

3. Prototype Development

An initial working model of the system is developed.

4. User Evaluation

Users interact with the prototype and provide feedback.

5. Prototype Refinement

The prototype is improved according to user suggestions and requirements.

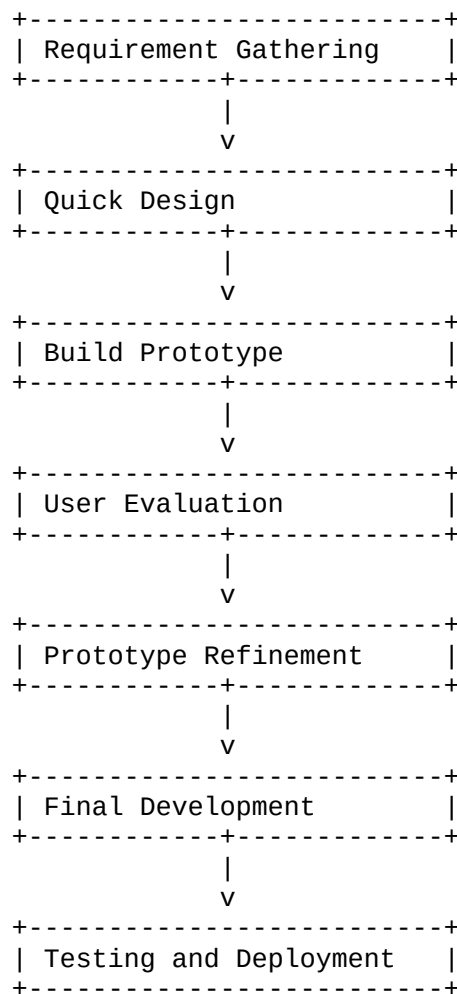
6. Final System Development

The complete system is developed after requirements are finalized.

7. Testing and Deployment

The final system is tested for errors and deployed for practical use.

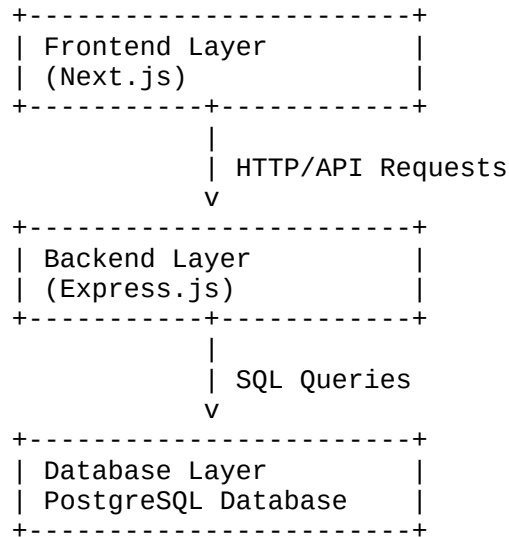
2.4 Prototyping Model Diagram



3. System Architecture

3.1 Overall Architecture

The Information System for Mart follows a three-layer architecture consisting of the frontend layer, backend layer, and database layer.



This architecture helps maintain separation between user interface, business logic, and data management.

3.2 Frontend Layer

The frontend layer is responsible for user interaction and interface rendering. It provides pages for login, inventory management, product searching, sales recording, and dashboards.

Technologies Used

- Next.js
- HTML
- CSS
- JavaScript

Responsibilities

- Display user interface
- Handle user input
- Send requests to backend APIs
- Display reports and analytics

3.3 Backend Layer

The backend layer handles system logic, API requests, authentication, and communication with the database.

Technologies Used

- Node.js
- Express.js

Responsibilities

- Process user requests
- Handle authentication
- Manage inventory logic
- Record sales transactions
- Calculate profit
- Manage role-based access control

3.4 Database Layer

The database layer stores all application data including users, products, sales records, and inventory information.

Technology Used

- PostgreSQL

Responsibilities

- Store user data
- Store product information
- Maintain sales records
- Handle stock updates
- Support business analytics

4. Module Design

4.1 Authentication Module

The authentication module provides secure access to the system.

Main Functionalities

- User login
- Password validation
- Role-based access control
- Session management

User Types

- Owner
- Worker

4.2 Inventory Management Module

The inventory management module handles all product-related operations.

Main Functionalities

- Add new products
- Update product details
- Delete products
- Manage stock quantity
- Assign rack and shelf locations
- Categorize products

This module ensures accurate and organized inventory tracking.

4.3 Product Search Module

The product search module allows users to locate products quickly.

Main Functionalities

- Search by product name
- Search by SKU
- Search by category
- Display rack and shelf location

This module improves efficiency during daily store operations.

4.4 Sales Management Module

The sales management module records sales transactions and updates inventory automatically.

Main Functionalities

- Record product sales
- Update stock in real time
- Prevent negative inventory
- Store transaction history
- Calculate transaction profit

This module helps maintain accurate sales and inventory records.

4.5 Reporting and Analytics Module

The reporting module provides business insights through reports and dashboards.

Main Functionalities

- Daily sales reports
- Profit analysis
- Product performance reports
- Inventory status reports
- Dashboard analytics

This module supports better business decision-making.

5. Database Design

5.1 Users Table

Field Name	Data Type
id	UUID
username	VARCHAR
password	TEXT
role	VARCHAR

Purpose

Stores login credentials and user roles.

5.2 Products Table

Field Name	Data Type
id	UUID
name	VARCHAR
sku	VARCHAR
category	VARCHAR
stock_count	INTEGER
rack_number	INTEGER
shelf_level	VARCHAR
cost_price	DECIMAL
selling_price	DECIMAL

Purpose

Stores inventory and product-related information.

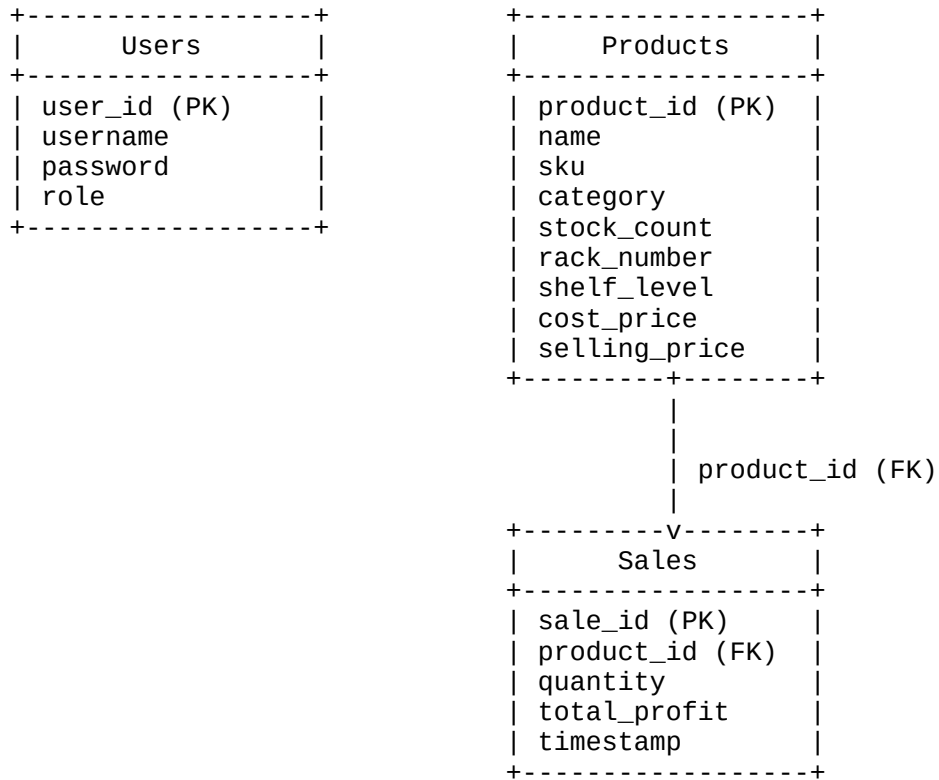
5.3 Sales Table

Field Name	Data Type
id	UUID
product_id	UUID
quantity	INTEGER
total_profit	DECIMAL
timestamp	TIMESTAMP

Purpose

Stores sales transaction records.

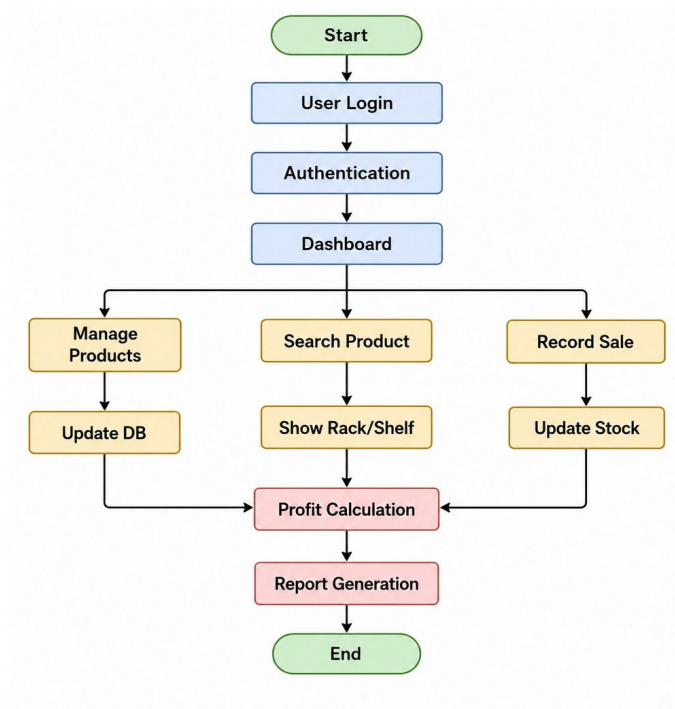
5.4 Database Relationship Overview



The sales table is connected to the products table using the product_id field.

6. Diagrams and Flowcharts

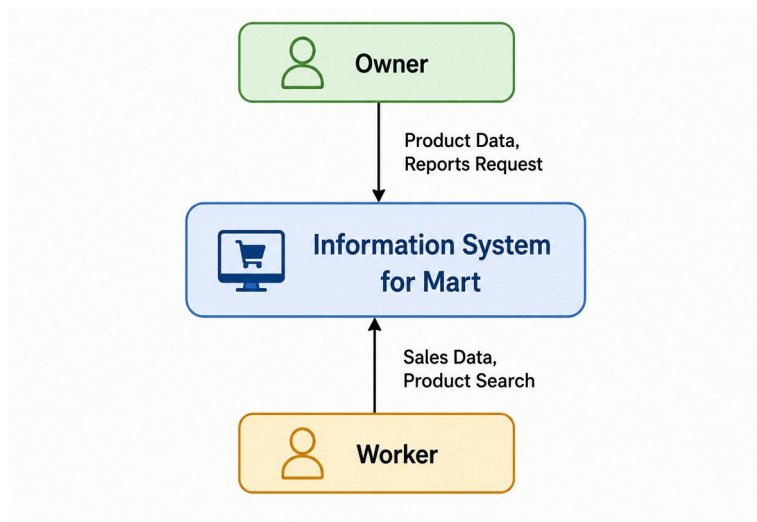
6.1 System Flowchart



6.2 Data Flow Diagram (DFD)

6.2.1 Context Level DFD (Level 0)

The Context Level Data Flow Diagram represents the overall interaction between users and the Information System for Mart.



Description

- The Owner manages inventory, reports, and analytics.
- The Worker performs sales and product searches.

- The system processes all requests and maintains records in the database.

6.2.2 Level 1 DFD

The Level 1 DFD divides the system into major functional modules.



Description

- Authentication Module verifies users.
- Inventory Module manages products and stock.
- Product Search Module locates products efficiently.
- Sales Module records transactions and updates inventory.
- Reporting Module generates reports and analytics.
- All modules interact with the PostgreSQL database.

6.3 Data Flow Description

1. The user logs into the system.
2. Authentication is verified through the backend.
3. The user accesses the dashboard.
4. Product information is retrieved from the database.
5. Sales transactions are recorded and inventory is updated.

6. Profit calculations are generated automatically.
7. Reports and analytics are displayed to the user.

7. User Interface Design

7.1 Login Page

The login page allows users to securely access the system using their username and password.

Main Components

- Username field
- Password field
- Login button
- Error message display

7.2 Dashboard Page

The dashboard provides an overview of store activities.

Main Components

- Daily sales summary
- Profit overview
- Inventory statistics
- Navigation menu
- Recent transactions

7.3 Inventory Page

The inventory page is used to manage products and stock.

Main Components

- Product table
- Add product form
- Update product option
- Delete product option
- Rack and shelf details

7.4 Sales Page

The sales page allows workers to record transactions.

Main Components

- Product search bar
- Quantity selection
- Sales confirmation button
- Transaction history

8. Technology Stack

Layer	Technology
Frontend	React.js
Backend	Node.js + Express.js
Database	PostgreSQL
Authentication	JWT
Version Control	Git & GitHub
API Testing	Postman
Development Tool	VS Code

9. Advantages of Proposed System

The proposed system provides several advantages compared to traditional manual store management systems.

Advantages Include

- Faster inventory management
- Improved product organization
- Accurate stock tracking
- Reduced manual calculation errors
- Better security and access control
- Faster sales processing
- Real-time inventory updates
- Improved business monitoring

- Better decision-making through analytics
- Scalable system architecture

10. Future Enhancements

The system can be improved further in future versions by introducing advanced features.

Possible Future Enhancements

- Barcode scanner integration
- AI-based sales prediction
- Mobile application support
- Cloud backup system
- Multi-store management
- Automated invoice generation
- Low stock notifications
- Supplier management system
- Advanced analytics dashboard
- Customer purchase analytics

11. Conclusion

The Initial Design of the Information System for Mart presents the proposed architecture, modules, database structure, and development methodology for the system. The design focuses on building an efficient, scalable, and user-friendly platform for retail business management.

The use of the Prototyping Model ensures continuous user involvement and system improvement during development. Through modules such as inventory management, sales processing, product searching, and analytics reporting, the system aims to simplify retail operations and improve business efficiency.

Overall, the proposed design provides a strong foundation for developing a practical and reliable retail management solution capable of supporting future enhancements and business growth.